

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think>
First, I will find out the time taken to download 60% of the file,
Since, speed = 2 GB / minute ,
 $60\% \text{ of } 200 \text{ GB} = 0.6 * 200 = 120 \text{ GB}$,
 $\text{Time taken} = 120 \text{ GB} / 2 \text{ GB / minute} = 60 \text{ minutes}$,
After 60 minutes , Windows forces a restart ,
Now , this 60 minutes are added to the total time ,
So , total time so far = 60 minutes ,
Now , Carla has to download 200 GB again ,
 $\text{So , time taken to download } 200 \text{ GB} = 200 \text{ GB} / 2 \text{ GB / minute} = 100 \text{ minutes}$,
 $\text{Total time taken} = 60 \text{ minutes} + 100 \text{ minutes} = 160 \text{ minutes}$,
</think> <answer>160 </answer>

